RONSM
EDUCATIONAL SERVICES

Follow our whats app channel

Click HERE

MOCK PREPARATION SETS

A LEVEL (UACE)

SUBJECT: PHYSICS

SET: Set 1 of 10

FACILITATOR: Nsereko Robert

0750661240

For the guide contact 0750661248

SET 1

SYLLABUS ALIGNMENT MAP

This table helps a teacher quickly find which scenario to use for their current lesson.

Item	Element of construct	Topics covered
Section A(compulsory)		
Item 1 **Force and motion**	Analysis of how force affects the state of bodies	➢ Measurement and Dimensions of Physical Quantities ➢ Statics ➢ Linear Motion ➢ Motion Under Gravity ➢ Work, Energy and Power ➢ Solid Friction ➢ Fluid Mechanics ➢ Mechanical Properties of Matter ➢ Circular Motion ➢ Gravitation
Item two(2) **Particles**	Investigation of atomic, quantum, and nuclear phenomena and their applications	➢ Atomic Particles ➢ Quantum Theory ➢ Nuclear Processes
Section B (part I) Attempt one question only		
Item three and four(3 & 4) **Energy**	Exploration and application of energy forms and transformations.	➢ Thermometry ➢ Heat Quantities ➢ Transfer of Heat ➢ Behaviour of Gases ➢ Thermodynamics

		➢ Reflection of Light ➢ Refraction of Light ➢ Optical Instruments ➢ Simple Harmonic Motion ➢ Progressive Waves ➢ Stationary Waves ➢ Sound Waves
Section B (part II) Attempt one question only		
Item five and six (5 & 6) **Charges and fields**	Exploration of how electric and magnetic fields interact in systems and devices.	➢ Electrostatics ➢ Capacitors ➢ Digital Electronics ➢ Current Electricity ➢ Magnetism in Matter ➢ Magnetic Effect of an electric Current ➢ Electromagnetic Induction ➢ A.C Circuits

UPDATED A LEVEL NEW CURRICULUM PHYSICS PAPER STRUCTURE

INSTRUCTIONS TO CANDIDATES:

- ✓ This examination paper consists of two sections; A and B. It has six items.
- ✓ Section A has two compulsory items.
- ✓ Section B has two parts; I and II. Respond to only one item from each part.
- ✓ Respond to four items in all.
- ✓ Any additional item(s) responded to, will not be scored.
- ✓ Responses to each item should start on a fresh page.
- ✓ Silent non-programmable scientific calculators may be used.

FORCE AND MOTION

Item 1

A commercial agribusiness is designing a chute system to transport harvested tomatoes from a higher sorting platform to a lower packaging area. The wooden chute is inclined at an angle of 25° to the horizontal. A crate full of tomatoes has a total mass of 40 kg. To prevent bruising the tomatoes, the quality control manager specifies that the crate must not accelerate down the chute at more than 1.2 m/s^2. The coefficient of kinetic friction between the crate and the chute is 0.18. A worker controls the descent by pulling on a rope attached to the crate, parallel to the chute. (Take g = 9.81 m/s^2).

Tasks:

(a) Calculate the required tension in the rope to ensure the crate accelerates at exactly 1.2 m/s^2. If the maximum continuous pulling force the average worker can safely exert is 100 N, evaluate whether this manual system is viable or if mechanical intervention is required.

(b) For irrigation, the agribusiness uses a water pump connected to a horizontal pipe that narrows significantly at the outlet. Explain, using relevant physical principles, why the speed of the water increases at the narrow outlet and how the fluid pressure changes in that region.

Item 2

Architects and structural engineers are collaborating on a restorative housing settlement in the Kawuku-Kakindu wetland. To preserve the natural water flow, the houses are elevated on solid, cylindrical steel support columns. One such column has an unstretched length of 3.5 m and a radius of 0.06 m. It supports a static structural load of 18,000 kg from the building above. The physical planning guidelines dictate that to maintain the precise G-Floor building levels, the vertical compression of any single column must not exceed 0.04 mm.

Hint

- Young's modulus of the steel = 2.0×10^{11} Pa;
- g = 9.81 m/s².

Tasks:

(a) Determine the compressive strain and the absolute compression (extension) of the steel column. Use your obtained values to advise the engineers on whether the current column design satisfies the physical planning guidelines for the G-Floor levels.

(b) During a severe weather event, the structural columns may experience sudden, extreme forces. Explain, using the concept of elasticity, the difference between elastic and plastic deformation, and discuss why it is critical that the support columns do not exceed their elastic limit.

Item 3

An electric mobility company is testing a new electric bicycle designed for navigating steep, winding rural roads. The combined mass of the test rider and the e-bicycle is 115 kg. During a test run, the rider approaches an unbanked,

horizontal corner with a radius of curvature of 18 m. Due to loose gravel, the coefficient of static friction between the tires and the road surface is reduced to 0.35. (Take g = 9.81 m/s²).

Tasks:

(a) Calculate the maximum safe speed (in km/h) at which the rider can negotiate the curve without skidding outward. If the e-bike's speedometer reads 30 km/h as they enter the corner, use your value to advise if a crash is imminent.

PARTICLES

Item 4

To protect a high-value poultry feed formulation plant, a security firm installs an advanced photoelectric intrusion detection system along the perimeter fence. The alarm circuit relies on a continuous beam of light striking a specialized metallic cathode made of Potassium, which has a work function of 2.30 eV. If an intruder breaks the light beam, the continuous emission of photoelectrons stops, instantly triggering the alarm relay. The night-time illumination is provided by a hidden ultraviolet LED array emitting monochromatic light at a wavelength of 350 nm.

(Planck's constant, h = 6.63×10^{-34} J s; Speed of light, c = 3.0×10^{8} m/s; Electronic charge, e = 1.6×10^{-19} C; Mass of an electron, me = 9.11×10^{-31} kg).

Tasks:

(a) By determining the energy of the incident ultraviolet photons, calculate the maximum kinetic energy (in Joules) and the maximum velocity of the emitted photoelectrons. Furthermore, compute the threshold frequency of the Potassium cathode and the required stopping potential of the security circuit.

(b) During maintenance, a technician accidentally replaces the 350 nm UV LED with a standard red LED emitting at 700 nm. Using Einstein's quantum

theory of light, explain why classical wave theory fails to predict the outcome of this change, and mathematically justify whether the alarm system will still function properly with the red LED.

Item 5

A medical physicist at a regional referral hospital is testing a newly installed Coolidge X-ray tube for diagnostic radiography. The tube is operated at an accelerating potential difference of 85 kV with an electron beam current of 15 mA. To ensure the longevity of the tungsten target (atomic number Z = 74), a continuous closed-loop oil cooling system is used. During calibration, it is determined that exactly 99.0% of the kinetic energy of the incident electrons is converted into thermal energy at the anode, while only 1.0% is successfully converted into X-ray photons.

Hint

- Planck's constant, $h = 6.63 \times 10^{-34}$ J s
- $c = 3.0 \times 10^{8}$ m/s
- $e = 1.6 \times 10^{-19}$ C

Tasks:

(a) Calculate the total input electrical power supplied to the X-ray tube and determine the exact rate of heat generation (in Joules per second) that the oil cooling system must dissipate to prevent the tungsten target from melting. Furthermore, calculate the minimum wavelength (cut-off wavelength) of the continuous X-ray spectrum emitted.

(b) The radiographer needs to image a patient with a much denser, thicker chest wall and therefore requires a more highly penetrating 'hard' X-ray beam. Explain the fundamental physical difference between continuous and characteristic X-rays. Outline the specific adjustments the radiographer must make to the control console to increase the penetrating power (quality) and the intensity (quantity) of the beam.

Item 6

An agricultural research institute is utilizing a radioactive tracer, Phosphorus-32 (P-32), to study the real-time uptake of liquid fertilizers in a ¾ acre experimental tomato farm. Phosphorus-32 decays by emitting beta-minus (β^-) particles and has a half-life of 14.3 days. A concentrated sample of the traced fertilizer with an initial radioactive activity of 4.0×10^5 Bq is injected into the farm's drip irrigation system. Public health regulations mandate that the residual radioactivity in the tomato tissues must fall below a safe consumption threshold of 500 Bq before the harvest can be safely distributed to local markets.

Tasks:

(a) Write the balanced nuclear decay equation for Phosphorus-32 (Atomic number of P = 15; Sulfur, S = 16). Calculate the decay constant of the isotope (in $days^{-1}$), and determine the exact number of days the researchers must wait before the tomatoes reach the safe consumption threshold.

(b) Explain the sub-atomic mechanism of beta-minus (β^-) decay in terms of nucleon transformation. Given the ionizing nature of beta particles, discuss two specific safety precautions the agricultural researchers must employ when initially handling and injecting the concentrated P-32 fertilizer in the field.

ENERGY

Item 7

A poultry feed manufacturing unit utilizes an industrial boiler to sterilize wet feed mixtures. The boiler contains 50.0 kg of water initially at an ambient temperature of 25.0°C. An electric heating element rated at 25.0 kW is switched on to heat the water to its boiling point of 100.0°C and subsequently vaporize exactly 15.0 kg of

it into steam to maintain high-pressure sterilization. The plant manager requires the entire heating and vaporization process for a single batch to be completed in under 35.0 minutes to meet production targets. Assume no heat is lost to the surroundings.

(Specific heat capacity of water = 4200 J kg^{-1} K^{-1}; Specific latent heat of vaporization of water = 2.26 × 10^6 J kg^{-1}).

Tasks:

(a) Calculate the total thermal energy required for this sterilization cycle and determine the exact time (in minutes) the electric heater must operate. Use your obtained value to advise the plant manager on whether the current 25.0 kW heater satisfies the 35.0-minute production target or if an upgrade is required.

(b) During maintenance, an engineer opens a pressure release valve, allowing some hot steam to expand rapidly (adiabatically) into the cooler atmosphere. State the First Law of Thermodynamics, and use it to explain the exact physical mechanism that causes the rapidly expanding steam to experience a sudden drop in temperature.

Item 8

An agricultural botanist is using a laboratory compound microscope to inspect tomato leaf cell samples for a newly identified fungal spore. The microscope features an objective lens of focal length 2.0 cm and an eyepiece lens of focal length 5.0 cm. The botanist places the illuminated tomato leaf slide exactly 2.5 cm from the optical center of the objective lens. The instrument is adjusted so that the final virtual image is formed at the standard near point (least distance of distinct vision), which is 25.0 cm from the eyepiece. To properly identify the microscopic fungal spores, the total angular magnification of the microscope must exceed 20×.

Tasks:

(a) Calculate the image distance for the objective lens and the object distance for the eyepiece lens. Subsequently, determine the total magnifying power of the compound microscope. Use your final value to evaluate whether the instrument meets the minimum 20× magnification requirement for spore identification.

(b) While viewing the highly magnified image, the botanist notices that the edges of the tomato cell walls exhibit colored fringes, distorting the fine details. Identify this specific optical defect. Explain the physical mechanism behind its occurrence in convex lenses and describe how the manufacturer can eliminate it using a compound lens design.

Item 9

During the structural testing of a restorative housing settlement built on the Kawuku-Kakindu wetland, engineers evaluate the stability of a suspended ground floor section against vertical wind gusts. The 400.0 kg floor section is supported by heavy-duty industrial springs. When the floor is initially mounted, it compresses the springs downward by a static distance of 0.05 m until reaching equilibrium. A severe simulated wind gust then displaces the floor vertically downwards by an additional 0.02 m from its equilibrium position before abruptly stopping, leaving the floor to oscillate freely. Safety guidelines state that to prevent structural cracking of the rigid timber cladding, the maximum oscillating velocity of the floor must never exceed 0.35 m/s. (Take $g = 9.81\ m/s^2$).

Tasks:

(a) Determine the effective force constant of the spring system, the angular frequency of the resulting oscillations, and the maximum velocity achieved by the floor section. Use your computed velocity to advise the engineering team whether the current suspension system complies with the structural safety guidelines for the timber cladding.

(b) The engineers decide to install hydraulic dampers to safely control these vibrations. Define damping, and explain the continuous energy

transformations taking place during one complete undamped oscillation cycle. Conclude by advising the engineers on which specific type of damping (underdamping, overdamping, or critical damping) is most appropriate for this wetland house, justifying your choice.

CHARGES AND FIELDS

Item 10

A commercial agribusiness center near Mukono is setting up an off-grid solar DC mini-grid to power a specialized poultry feed milling machine. The primary battery bank provides a stable electromotive force (e.m.f.) of 48.0 V with an internal resistance of 0.05 Ω. The milling machine motor is situated at a distribution distance of 150 m from the battery bank. Power is transmitted via a two-core copper cable where each core conductor has a cross-sectional area of 6.0×10^{-6} m^2. The milling machine motor requires a minimum terminal voltage of 42.0 V to start up and operate safely without drawing excessive current, which would otherwise burn out its internal windings. When running at full load, the motor has an effective resistance of 1.20 Ω. (Resistivity of copper at operating temperature = 1.68×10^{-8} Ω m).

Tasks:

(a) Calculate the total loop resistance of the transmission cable conductors, the total current flowing through the network during full-load operations, and the actual terminal voltage delivered to the milling motor. Based on your calculated voltage, advise the plant operators whether the system satisfies the safe startup parameters or if thicker transmission cables must be installed.

(b) Over long periods of continuous operation, the distribution cables experience a significant temperature rise. Explain, using a microscopic model of electrical conduction, why the electrical resistance of metallic conductors increases with temperature, and discuss the economic and technical consequences of this effect on power distribution efficiency.

Item 11

An innovative e-mobility workshop in Kampala is testing a prototype regenerative braking storage module for electric bicycles. The system captures kinetic recovery energy during downhill descents and stores it temporarily in a bank of parallel-plate capacitors. A single experimental unit consists of two large aluminum sheets, each of surface area 0.85 m^2, separated by a thin dielectric sheet of mica of thickness 0.25 mm. The mica layer possesses a relative permittivity (dielectric constant) of 6.0. The backup assist circuit charges this capacitor assembly using a step-up charging potential of 12.0 V. To provide a viable power torque assist to the rider climbing steep terrain, each capacitor cell must be able to release a minimum electrostatic stored energy of 2.0×10^{-6} J. (Permittivity of free space, $\varepsilon_o = 8.85 \times 10^{-12}$ F/m).

Tasks:

(a) Determine the capacitance of the experimental storage cell, the total charge accumulated on its plates when fully charged, and the total electrostatic energy stored within its field. Use your computed energy value to evaluate whether this parallel-plate configuration meets the necessary torque-assist energy criteria specified by the workshop.

(b) During field testing under humid conditions, a puncture occurs in the dielectric material, creating a tiny air gap. Explain, from a molecular perspective, how the presence of a polarized dielectric material fundamentally increases capacitance compared to a vacuum or an air gap, and explain the physical meaning of the term dielectric breakdown strength.

Item 12

A micro-hydroelectric plant along a fast-flowing river in Kapchorwa utilizes a localized synchronous generator to provide power to neighboring households. The armature consists of a flat, rectangular coil containing 450 turns of copper wire, with dimensions of 20 cm by 15 cm. The turbine mechanism forces this coil to

rotate at a uniform rate of 1500 revolutions per minute (rpm) inside a uniform magnetic field of flux density 0.12 T produced by fixed permanent magnets. To achieve stable grid integration with small consumer household electronics, the peak induced electromotive force (e.m.f.) from the armature assembly must hit a minimum target voltage of 240 V.

Tasks:

(a) Formulate the expression for the total magnetic flux linkage through the generator coil as a function of time. Compute the precise peak electromotive force (e.m.f.) produced by this generator. Based on your computed value, advise the local planning committee whether the generator satisfies national standard consumer parameters or if the mechanical gearing ratio must be altered to increase turbine angular velocity.

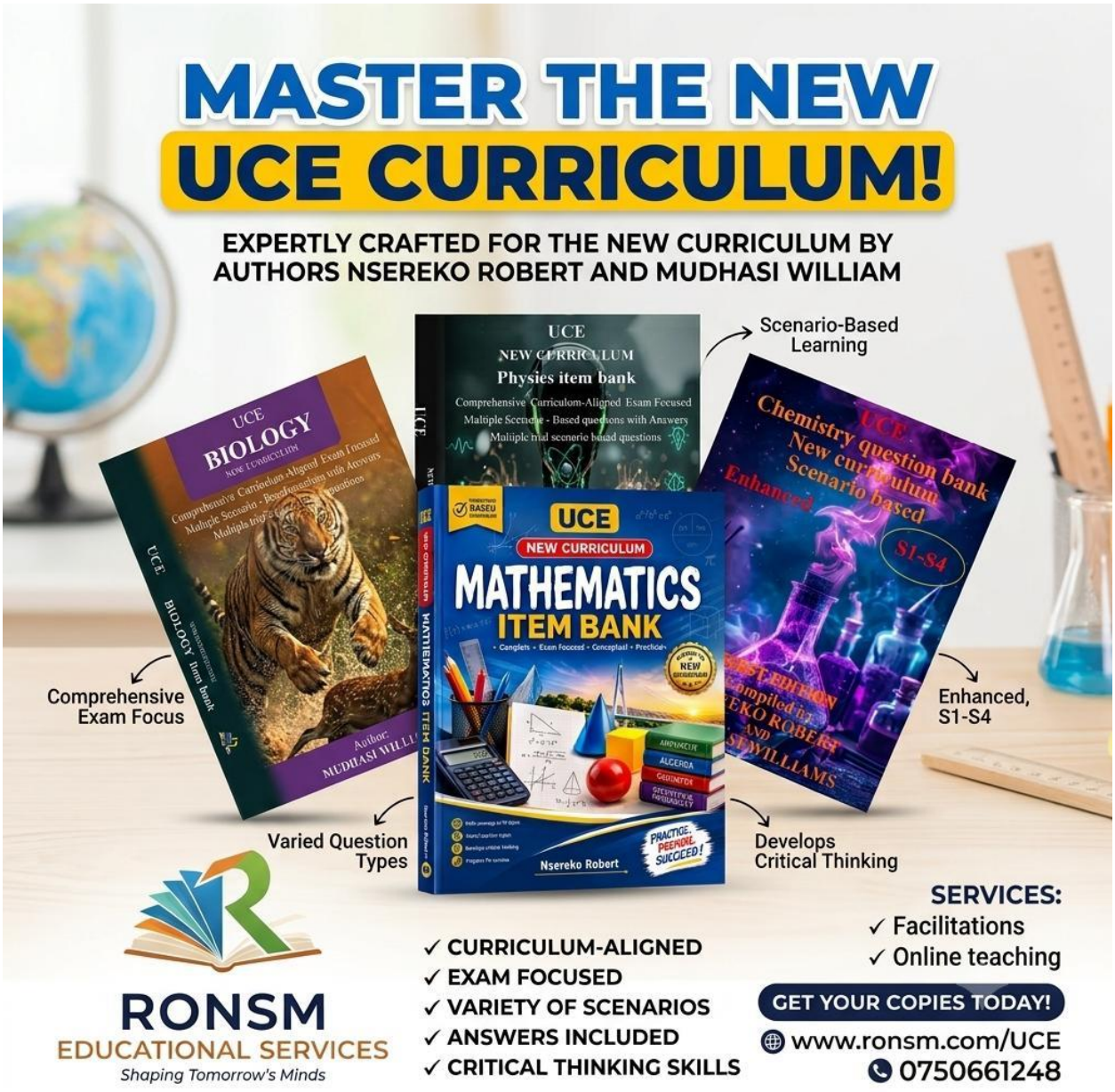
(b) When a high electrical load is suddenly switched on by multiple households, the turbine experiences an immediate mechanical deceleration. State Lenz's law, and explain the physical mechanisms responsible for this back-torque effect, outlining the interaction between the induced armature current and the external magnetic field.

End

Nurturing future engineers and scientists

Follow our whats app channel

Click HERE


MASTER THE NEW
UCE CURRICULUM!
EXPERTLY CRAFTED FOR THE NEW CURRICULUM BY AUTHORS NSEREKO ROBERT AND MUDHASI WILLIAM
Scenario-Based Learning
Comprehensive Exam Focus
Enhanced, S1-S4
Varied Question Types
Develops Critical Thinking
UCE
NEW CURRICULUM
Physics item bank
UCE
BIOLOGY
UCE
NEW CURRICULUM
MATHEMATICS
ITEM BANK
Nsereko Robert
PRACTICE. PERFORM. SUCCEED!
UCE
Chemistry question bank
New curriculum
Scenario based
Enhanced
S1-S4
RONSM
EDUCATIONAL SERVICES
Shaping Tomorrow's Minds
✓ CURRICULUM-ALIGNED
✓ EXAM FOCUSED
✓ VARIETY OF SCENARIOS
✓ ANSWERS INCLUDED
✓ CRITICAL THINKING SKILLS
SERVICES:
✓ Facilitations
✓ Online teaching
GET YOUR COPIES TODAY!
www.ronsm.com/UCE
0750661248

Prepared by;

NSEREKO ROBERT

0750661248

RONSM Educational services

Other products offered by RONSM Educational services;

Item banks

- Physics
- Biology
- Chemistry
- Mathematics

Online lessons.

Facilitations

Contact 0750661248

Embracing academic excellence

Follow our whats app channel

Click HERE

Nsereko Robert 0750661248 RONSM Educational services



act 0750661248